



Updating Beliefs with Evidence

Bayes' Theorem from First Principles to Modern Practice

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Trial Lecture

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Goals for Today

1

Get familiar

Build a working command of probability and Bayes' theorem.

2

Build intuition

Develop a feel for how evidence updates belief — beyond the formulas.

3

See it applied

Walk through practical applications of Bayesian inference in the real world.

4

Hands-on practice

Implement Bayesian inference yourself, end-to-end.

Motivation • The Mammogram Paradox

A 40-year-old woman tests positive on a mammogram. The test has 90% sensitivity and 91% specificity. Breast cancer prevalence at this age is about 1%. For simplicity, we assume a population of 1,000,000.

What is the probability she actually has cancer?

- (a) ~90% (b) ~75% (c) ~50% (d) ~9%

We'll return to this answer in a few minutes.



PART 1

Foundations

From conditional probability to Bayes' theorem



1.1 • Conditional Probability

$$P(A \mid B) = P(A \cap B) / P(B)$$

"The probability of A, given that B has occurred."

- **Crucially: $P(A \mid B) \neq P(B \mid A)$**
- $P(\text{positive test} \mid \text{cancer})$ tells us about the test
- $P(\text{cancer} \mid \text{positive test})$ is what the patient wants to know

This asymmetry is the source of the mammogram paradox.

Joint probabilities of events A and B (rows and columns sum to marginals)

	B	not B	Total
A	$P(A \cap B)$	$P(A \cap B^c)$	$P(A)$
not A	$P(A^c \cap B)$	$P(A^c \cap B^c)$	$P(A^c)$
Total	$P(B)$	$P(B^c)$	1

$$P(A \mid B) = P(A \cap B) / P(B)$$

numerator = joint cell · denominator = column marginal

1.2 • Deriving Bayes' Theorem

Start from one identity. The rest is algebra.

Step 1. $P(A \cap B) = P(A | B) \cdot P(B)$

Step 2. $P(A \cap B) = P(B | A) \cdot P(A)$

Step 3. Set them equal, divide by $P(B)$:

$$P(A | B) \cdot P(B) = P(B | A) \cdot P(A)$$

$$P(A | B) = P(B | A) \cdot P(A) / P(B)$$

1.3 • Naming the Parts

$$P(A \mid B) = P(B \mid A) \cdot P(A) / P(B)$$

1.3 • Naming the Parts

$$P(H | E) = P(E | H) \cdot P(H) / P(E)$$

Prior $P(H)$ — what we believed before seeing E

Likelihood $P(E | H)$ — how well H explains the evidence

Posterior $P(H | E)$ — what we believe after seeing E

Evidence $P(E)$ — the normalizing constant



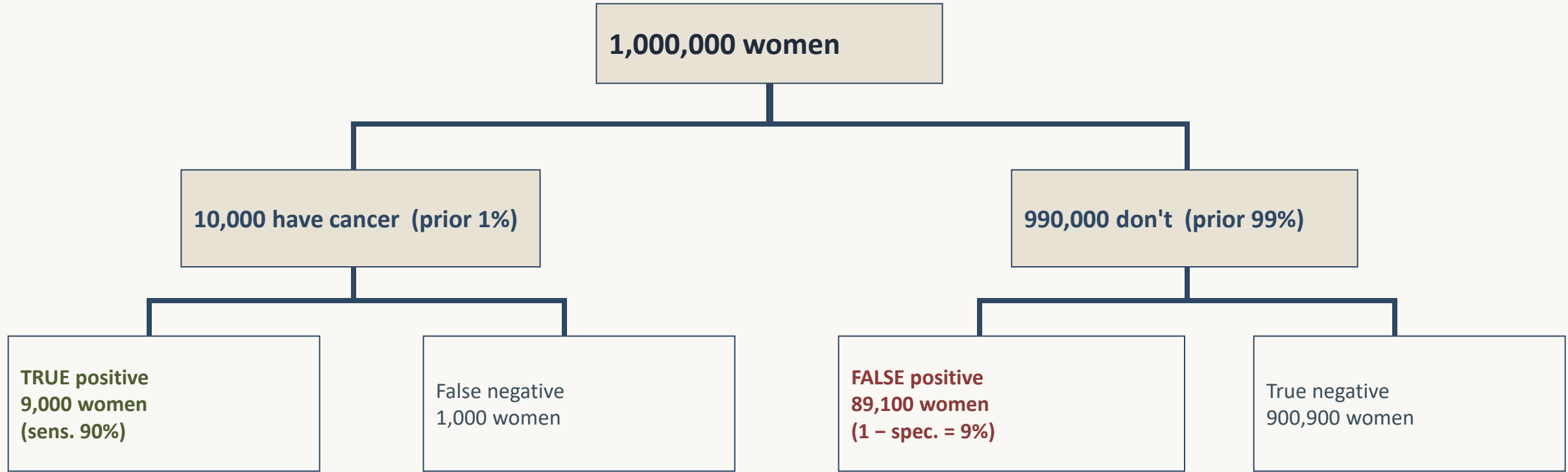
PART 2

Solving the Paradox

Actual numbers and the base-rate insight



2.1 • Solving the Paradox – In Terms of People



Among $9,000 + 89,100 = 98,100$ positive tests, only 9,000 are true cancers $\rightarrow P(H|E) \approx 9.2\%$

2.2 • Solving the Paradox – Via Bayes' Theorem

Let H = "has cancer", E = "positive test". We want $P(H | E)$.

Bayes' theorem:

$$P(H | E) = \frac{P(E | H) \cdot P(H)}{P(E)}$$

2.2 • Solving the Paradox – Via Bayes' Theorem

Let H = "has cancer", E = "positive test". We want $P(H | E)$.

Bayes' theorem:

$$P(H | E) = \frac{P(E | H) \cdot P(H)}{P(E | H) \cdot P(H) + P(E | \neg H) \cdot P(\neg H)}$$

Plug in:

$$P(E | H) = 0.90$$

sensitivity

$$P(H) = 0.01$$

prevalence

$$P(E | \neg H) = 0.09$$

1 – specificity

$$P(\neg H) = 0.99$$

no cancer (1-prevalence)

$$P(H | E) = (0.90 \times 0.01) / (0.90 \times 0.01 + 0.09 \times 0.99) = 0.009 / 0.0981$$

$\approx 9.2\%$ — same result as the natural-frequency picture

2.2 • The Base-Rate Insight

$$\text{posterior} \propto \text{likelihood} \times \text{prior}$$

- When the base-rate (prior) is small, even strong evidence (likelihood) yields a modest posterior.
 - The 1% prevalence dominates the 90% sensitivity — that's why the answer is ~9%.
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PART 3

Where Bayes Lives in Modern Practice



Where Bayes Lives in Modern Practice



Naïve Bayes

Spam classification —
"naïve" because it assumes
feature independence
given class.



Bayesian A/B

Credible intervals on
conversion lift — directly
answer "is B better than
A?"



Bayesian neural nets

Posteriors over network
weights → calibrated
uncertainty for safety-
critical predictions.



Medical diagnostics

Beyond mammograms —
diagnostic test
interpretation under
varying prevalence.

PART 4 • HANDS-ON

Bayesian A/B Test in Python

4.1 • Bayesian A/B Test in Python

Estimating the click-through rate θ of a new website button after k clicks in n trials.

Prior: $\theta \sim \text{Beta}(\alpha, \beta)$

Likelihood: $k \mid \theta \sim \text{Binomial}(n, \theta)$

Posterior: $\theta \mid k, n \sim \text{Beta}(\alpha + k, \beta + n - k)$

Conjugate prior $\rightarrow$ closed-form posterior. The hyperparameters update by simple addition: $\alpha \rightarrow \alpha + k$, $\beta \rightarrow \beta + n - k$.

4.2 • Bayesian A/B Test in Python

Notebook on [Christoph Metzner/Gisma-Trial-Lecture: Extra material for Gisma trial lecture - 12/5/2026](#) — *scipy.stats.beta* only, no heavy frameworks.

Scenario: Two website designs A and B. Conversion data provided. Should we ship B?

1. **Specify priors** — $\text{Beta}(1,1)$ uniform on each design's conversion rate
2. **Compute posteriors** — $\text{Beta}(1+k_A, 1+n_A - k_A)$ and similarly for B
3. **Estimate $P(B > A)$** — sample 10,000 draws from each posterior, compare

Stretch: *re-run with $\text{Beta}(50, 50)$ and observe how the posterior tightens. Sensitivity matters.*

CONCLUSION • Coming Back to Our Goals

01 • GET FAMILIAR

Bayes' theorem is the mathematics of learning from evidence — and now it's in your toolkit.

02 • BUILD INTUITION

Posterior $\propto$ likelihood $\times$ prior — when the base rate is small, even strong evidence yields a modest posterior.

03 • SEE IT APPLIED

The same framework powers Naïve Bayes, Bayesian A/B testing, Bayesian neural nets, and medical diagnostics.

04 • HANDS-ON PRACTICE

You've implemented it end-to-end — the rest is practice and curiosity.

CONCLUSION • **Material**

The material for the lecture can be found here:

[ChristophMetzner/Gisma-Trial-Lecture: Extra material for Gisma trial lecture - 12/5/2026](#)

Contains:

- **The lecture slides.**
 - **An interactive artifact for the mammogram paradox.**
 - **Another interactive example to help you build intuition on Bayes' theorem.**
 - **The Jupyter notebook for the hands-on Bayesian A/B testing example.**
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Q&A

"You started this session likely overestimating that probability. You now have a tool for not making that mistake — in medicine, in machine learning, in any decision under uncertainty."

Hands-on Notebook • Discussion Questions

With 1,000 visitors per arm, the data dominates the prior. **At what sample size** would you expect the prior choice to materially change the conclusion?

- The strong prior Beta(50, 450) carries a "pseudo-sample size" of $\alpha + \beta = 500$, and it acts like 500 prior observations centered at 10%. The prior materially competes with the data when the real sample size is on the same order or smaller. So:
 - At $n \approx 1,000$ per arm (current), data dominates and both priors give similar conclusions.
 - At $n \approx 100\text{--}200$ per arm, the strong prior pulls posteriors toward 10% enough to meaningfully shrink the gap between A and B, and $P(B > A)$ drops noticeably.
 - At $n \approx 50$ per arm, the strong prior essentially overrides the observed difference.

A useful rule of thumb: the prior matters when n is within an order of magnitude of $\alpha + \beta$.

Hands-on Notebook • Discussion Questions

Suppose Design B is more expensive to maintain. How might you incorporate that cost into the decision rule, beyond just checking $P(B > A) > 0.5$?

- $P(B > A) > 0.5$ only asks "is B better at all?" but it ignores *by how much*. Better approaches:
 - **Minimum effect threshold.** Require $P(\theta_B - \theta_A > \delta) > 0.95$, where δ is the lift needed to justify the extra cost (e.g., the conversion-rate increase that pays for the maintenance overhead).
 - **Expected value/loss.** Compute the posterior expected profit of each design: $E[\text{revenue_per_visitor} \times \theta - \text{cost}]$. Ship B only if its expected profit exceeds A's by a comfortable margin.
 - **Decision-theoretic loss function.** Define a loss for each (decision, true θ) pair, then pick the action minimizing expected posterior loss. This naturally penalizes shipping B when the lift is real but tiny.

The Bayesian posterior is the same in all cases; what changes is the decision rule applied on top of it.

Hands-on Notebook • Discussion Questions

The Beta-Binomial framework assumes each visitor's outcome is **independent and identically distributed**. When might that assumption fail in real-world A/B testing? (Think about repeat visitors, time-of-day effects, or marketing campaigns running in parallel.)

- Several common ways:
 - **Repeat visitors.** The same user converting (or not) twice isn't two independent trials; it's correlated. Cookie-based deduplication helps, but isn't perfect across devices.
 - **Time-of-day / day-of-week effects.** Conversion rates shift between weekday business hours and weekend evenings. If A and B aren't randomized at the user level *throughout* each period, you get confounding (the "weekend bug" problem).
 - **Parallel marketing campaigns.** A promotion or press mention during the test brings in a different traffic mix — visitors during that window aren't drawn from the same distribution as baseline visitors.
 - **Network/interference effects.** On social platforms, exposing one user to B can influence their friends (who may be in arm A), violating the stable-unit-treatment-value assumption.
 - **Novelty and primacy effects.** Returning users may click B simply because it's new; the lift decays once novelty wears off.
 - **Learning effects within a session.** A visitor's second pageview isn't independent of their first.

Mitigations include user-level (not visit-level) randomization, running tests for full weekly cycles, blocking on known confounders, and using hierarchical models that explicitly account for clustering.
